Anvesh Sharma

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EDUCATION

Boston University, Kilachand Honors College | GPA: 3.7 | Dean's List 2023–2025 **Boston, MA**
B.S. Biomedical Engineering, Robotics Concentration | Minor in Mechanical Engineering *Expected May 2027*
Relevant Coursework: Biomedical Instrumentation (graduate; LabVIEW), Intro to Robotics (Simulink), Solid Biomechanics (COMSOL), Device Design & Diagnostics, Thermodynamics

TECHNICAL SKILLS

Electronics & Test: Analog front-end design, filtering & signal conditioning, oscilloscope, function generator, DMM, optical comparator
Software: Python, MATLAB / Simulink, LabVIEW, Git / GitHub, Linux command line, Minitab, MasterControl
CAD & Design: SolidWorks, Fusion 360, FEA (COMSOL), GD&T, tolerance stack-up, DFM / DFA, engineering drawings
Prototyping: SLA/FDM 3D printing, manual mill, laser cutter, soldering, silicone molding, band saw, UV curing

ENGINEERING EXPERIENCE

AtriCure **Pleasanton, CA**
R&D Engineering Co-op *Jan 2026 – Aug 2026*
- Owned design and assembly of test fixtures for formal design verification of 8 cardiac ablation devices used with robotic surgical systems; data supported a 510(k) submission; released drawings in MasterControl and supported release of verification protocols
- Designed a modular SolidWorks/SLA-printed fixture on a Delrin base to test an ablation grasper in compression, bending, and torsion, targeting failure modes from an AFMEA
- Designed Instron fixtures to benchmark 24 hypotubes from 2 suppliers and test device designs to failure; analyzed results with normality and 2-sample t-tests in Minitab
- Investigated early LARIAT suture separation in a CAPA: reworked chamfered stepped tips into 0.004–0.006 in. fillets, cycled 3 modified devices to 100 cycles, and showed the chamfers were one cause of suture loop damage
- Presented root-cause analyses for field complaints; participated in design reviews and CAPA, FMEA, 510(k), and PRM discussions

Material Robotics Lab, Boston University **Boston, MA**
Undergraduate Researcher, advised by Dr. Sheila Russo & Viola Del Bono *Sep 2025 – Dec 2025*
- Designed and prototyped a low-cost 2D tactile sensor using a single optical waveguide for an artificial colon device; modeled parts and silicone molds in Fusion 360
- Programmed a Universal Robots arm in Python to apply set forces across sensor regions, automating data collection
- Processed sensor data in MATLAB and trained machine learning models to predict force and contact position from optical loss; checked data quality with Shapiro-Wilk, ANOVA, and t-tests

ELECTROMECHANICAL & ROBOTICS PROJECTS

LymphSense | *Team Lead* **Fall 2025 – Spring 2026**
- Led a student team designing a low-cost, handheld lymphedema detection device based on bioimpedance spectroscopy; the prototype detected Stage 2 lymphedema in collected patient data
- Redesigned the analog front end of an AD5933-based bioimpedance circuit, adding a voltage-follower buffer and Howland current pump for stable, load-independent current injection; switched to tetrapolar electrodes to minimize electrode–skin contact impedance, cutting measurement standard deviation by 50%
- BU BMES Design-a-Thon 1st Place & People's Choice; Dean's Imagineering Best Junior Team (2026); BMES 2026 poster presenter

Electromyogram (EMG) Light Sensor **Winter 2025**
- Designed a 3-electrode EMG device with an instrumentation amplifier, filtering and signal-conditioning stages, and an Arduino to detect nerve signal degradation; demonstrated proof of concept at the Lutron Light Competition

Line-Following Robot | *Team of 4* **Fall 2025**
- Implemented feedforward control with Dubins paths and LQR feedback control for obstacle avoidance on an AgileX Limo robot; validated paths with motion-capture data

Underwater Camera Drone | *BU Senior Design* **Fall 2026 – Present**
- Scoping requirements for an aquatic-life camera drone; designing its sealed chassis and motion-detection pipeline

LEADERSHIP & INVOLVEMENT

Society of Asian Scientists and Engineers (SASE) | *Mentor* **Fall 2026 – Present**

BU Rocket Propulsion Group, Hardware Team | *Subteam Lead* **Sep 2024 – Jan 2025**
- Led integration of the rocket nozzle with the solenoid valve subsystem; modeled parts in SolidWorks and used FEA and CAD design reviews to confirm the nozzle would withstand high-pressure combustion gas before build